HumanGenPhen Data Platform IS:
text mining resources for extracting genetic and phenotypic data from scientific literature

ELIXIR All Hands 2022, 7-10 June, Amsterdam, Netherlands

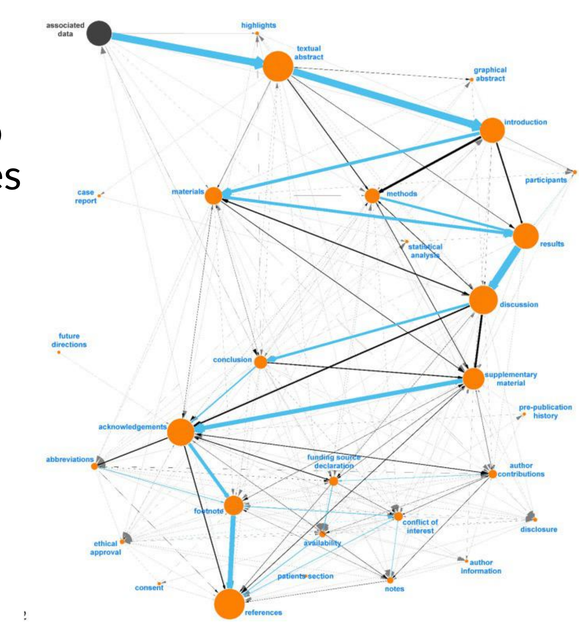
Health research is advanced through genome-wide association studies (GWAS) which provide a deeper understanding of disease aetiology by detecting associations between genetic variants and disease traits in population samples. New knowledge from GWAS is initially disseminated through the scientific literature and subsequently imported into databases in a process that relies on the efforts of biocurators to interpret individual publications. Text mining has been shown to facilitate accelerated biocuration and has a role in enabling the scalable extraction of human genotype and phenotype entities from biomedical literature.

The ELIXIR HumanGenPhen Data Platform implementation study is consolidating, developing, and using text mining tools and resources to support the automated curation of human genotype-phenotype associations in the GWAS Central and DisGeNET data services.

Tools for publication text and table standardisation with section classification

The Auto-CORPus package converts publication HTML into the BioC standard and annotates publication sections with Information Artifact Ontology (IAO) terms. Three output files are produced per publication:

- Full-text is converted to *BioC-JSON* format
- Publication tables are converted to *tables-JSON* format
- Abbreviations and full definitions are related in *abbreviations-JSON* format



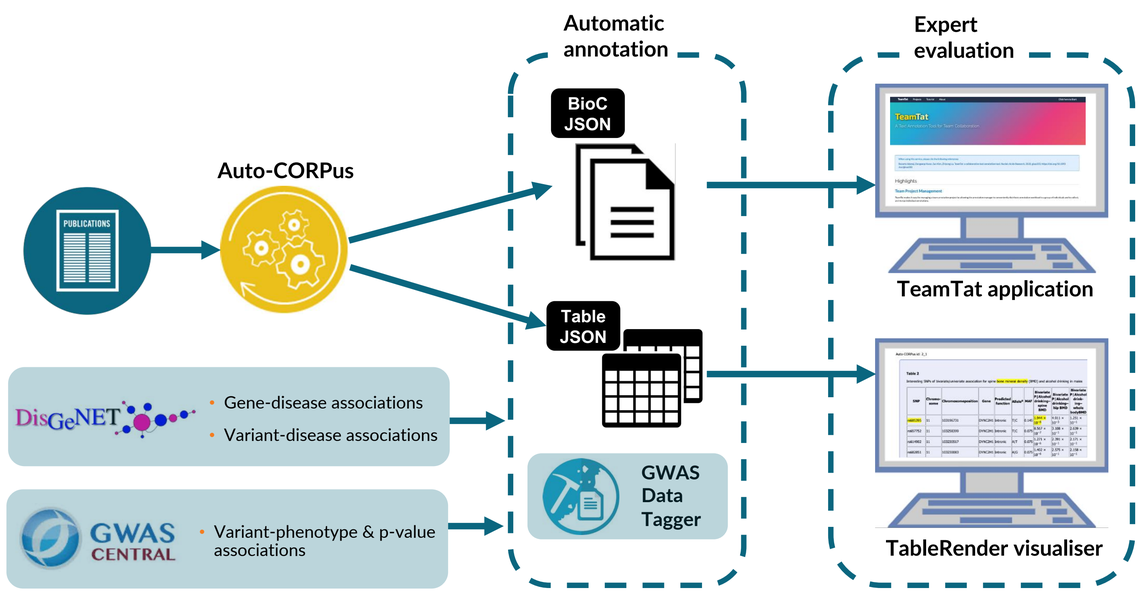
A directed graph is used by Auto-CORPus to assign the IAO publication section classification.

Figure taken from:
Beck T, et al. Front Digit Health. 2022;4:788124

Auto-CORPus available from:
https://github.com/omicsNLP/Auto-CORPus

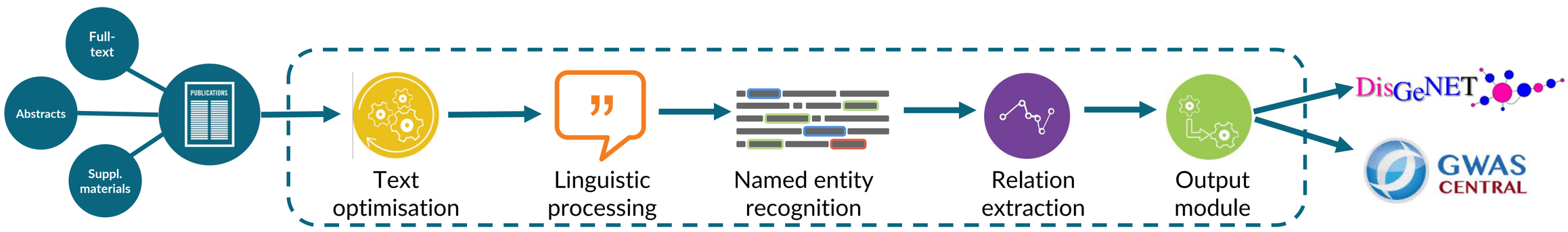
Annotated GWAS Corpus

We are building an annotated full-text article corpus for use in text mining research. The corpus is initially automatically processed by mapping GWAS Central and DisGeNET database content back onto the publications to annotate entities and relations (e.g., genetic variants, phenotypes and p-values) found in the full-text and tables. Data curators then review and, where necessary, extend the annotations.



Text mining workflow for genetic and phenotypic data extraction

We are developing a modular text mining workflow to extract genetic and phenotypic data from scientific literature. This workflow will be unique in its ability to analyse data sources that have previously been difficult to process, such as publication full-texts and supplementary materials. The workflow will be benchmarked against relevant corpora, including the annotated GWAS corpus described above. The workflow will convert publications into GWAS associations that can be loaded into DisGeNET and GWAS Central, hence supporting the efforts of biocurators and alleviating the requirements for manual interventions.



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